ActInf Livestream #015.1 ~ "Free-Energy Principle, Computationalism and

Livestream | 2:07:21

[Music]
hello everyone welcome to the active
inference lab
to the active inference live stream
today
is active inference live stream 15.1
and it is february 2nd 2021
welcome to the active inference lab
everyone we are an experiment
in online team communication learning
and practice
related to active inference you can find
us on our website
twitter discord gmail youtube or
keybase public team this is a recorded
and an archived live stream so please
provide us with
feedback so that we can improve on our
work all backgrounds and perspectives
are welcome here
and as far as video etiquette for live
streams we'll mute if there's noise in
our background
and raise our hands so we can hear from
everybody
today we are in 15.1 on february 2nd
and really excited and thankful to have
both authors of this paper
today and also joining us for next week
so that's going to be
really exciting check out this
spreadsheet if you want to
see the most updated event calendar for
active inference
live stream but here we are in 15 with a

rapid introduction
today we're gonna have a introduction to
each other especially some of
us who are new on the stream and then
we're going to be
kind of walking through the paper having
ideas having fun conversations and also
writing down ideas that we want to
discuss more in 15.2 which will be like
a follow-up
discussion on this so if you're
participating live or
watching later then just write down
ideas and questions you're having
and get in touch with us if you want to
participate
so here we are in the introductions
today we'll
go around introduce ourselves and then
just we're in it
so we'll have a fun conversation um i'm
daniel friedman i'm a postdoctoral
researcher
in california and yeah i'm just really
excited to be having this conversation
so i will pass it first to
our authors ines and thomas
up we can't hear you in s but you're not
muted so reload and
let's go to thomas
thomas can you unmute and introduce
yourself and then and that's just like
re
like control r reload and then you'll
probably work with the audio but go
ahead thomas
yeah um hi everyone my name is thomas
from

s um i'm
currently a phd student in
antwerp at the university of antwerp um
i'm working primarily on inactivism
and free energy principle
thanks for having me thank you we're
gonna look forward to hearing a lot more
from you and
s can we hear you now okay can you hear
me now perfect
great all right um all right so my name
is ines um i
am a postdoctoral fellow at the berlin
school of mind and brain
and yeah and my my topic of research is
on dynamical systems and
e-cognition in in general
great um then we can just pass it to
whoever hasn't spoken so
go to shannon
hey i'm shannon i'm a phd student at the
university of california in merced
um and i work on sensory motor
neuroscience and
the dynamics of crowd behavior um i'll
pass it to sasha
[Music]
i'm here let's go to steven then scott
david
oh hello i'm stephen i'm in toronto
i'm i work a lot with participatory
theater and community development i'm
doing a practice-based phd at the moment
through canterbury christ church
university and
i will pass it over to alex
hi everyone i'm alex i'm in moscow
russia

and i'm a researcher and systems
management school
and still trying to find ways to join
active influence
and systems thinking frameworks
and i pass it to session hi
um i'm sasha i'll try this again i'm a
graduate student in neuroscience
and studying early brain development
and based in california and i will pass
it to
blue good morning i am blue knight
i am an independent research consultant
based out of new mexico
i don't know who hasn't gone yet maybe
i'll pass it to marco
hi uh i'm marco marcos i'm based in
holland
um basically just a free energy active
influence enthusiast
um happy to be here i'll pass it to
scott
hi folks um my name is scott david i'm
the director of the information risk
research initiative at the university of
washington applied physics
lab my background is a lawyer and what
we're doing is engineering
systems that are socio-technical systems
so it's parameterization for technical
and
social elements of active inference
based systems
thanks well thanks everyone
so much for coming on live this is
really going to be a great
discussion and also a really prescient
paper so great

to hear everyone's views on it in these
warm-up questions
we can just begin where wherever people
have been
heading heading into today's discussion
from
so i'll just list them and then people
can raise their hand and
we'll hear from everybody who wants to
speak
what is something you're excited about
today what is something that you liked
or remembered about the paper
what motivated you to come to this
discussion to
read the paper think through these
topics and then also what's something
that you're
wondering about or would like to have
resolved by the end of today's or
next week's discussion so
until anyone raises their hand um ines
and or thomas um i just wanted to
ask how do you start by communicating
this
research or what led you to this
question
i'm going to kick out the ghost in esses
but um um yep so she should i take this
or do you wanna
but you're muted right now
sorry um yeah no thomas just go ahead
yeah
yeah um so i think the way
i got to the the idea of the paper is
that i'm
um primarily coming at it from a
radical inactivist perspective um

where there's a lot of
um antagonism towards
representations models and whatnot
um in the meantime i got very excited
about free energy principle and i've
been sort of trying to sort out where
that all fits what the models mean
what everything means in everything
that's being
said um and so ines and i were talking
about it
and um we started talking about
representations and models in the free
energy principle
and we figured out that
it really depends on how you interpret
the models and what exactly you take
the mathematics to mean um
[Music]
and based on that we just started
listing out the different options you
could have
and um what they mean and what we
personally thought about those what we
thought was
were good aspects less nice aspects
um and through all of that we
sort of came up with instrumentalism as
the uh
only path forward
awesome i'm sure we're gonna have a lot
of great
second thoughts and conversations on
that so anyone can
raise their hand but either an s or
thomas just to
lead into those who might not be
familiar with

long-running debates in the philosophy
of science or multiple possible
ways to interpret formalisms how do you
convey this instrumentalism realism
question and
motivate its importance
shall i take it thomas
sure okay um so
uh one thing that we thought was like
really interesting um to think about
is that um the majority
of the the accounts within the free
energy principle or process theories
uh they are very much realist so that's
what we
thomas and i started questioning and we
think that it's
it's interesting because uh there's been
other people also joining that
particular consensus that we also have
um
so we thought that this would be
something that we needed to look at
uh and the question just to sort of like
give a little bit of context on where we
are in the literature
is that the majority of the approach the
approaches were um on the realist camp
even those that were um coming from an
activism
and that's why we found things to be a
little bit strange because
the first question is whether one is
a licensed to take a realist account
from the
the fact that we can model uh the brain
or we can model the neural activity
or we can model um uh cognitive

cognitive activity
so that's the one question that we
wanted to to think about
and look at whether we are licensed to
take a realist stand and if we are what
is the argument what is the evidence to
think that
um this machinery that we we use in our
scientific model
modeling um can also be uh
exists somewhere in uh neuronal activity
or
cognitive activity so that was one thing
then the other thing was that it start
there started to be like a sort of like
a
wave that was taking um the free energy
to be
compatible with an activism and um
and it it it might we started thinking
about and questioning this because it
might be the case that
they may not be compatible after all
because as we know on the side of the
in activism uh there's like a claim
um against uh representations for
example
then on the predictive uh accounts of
cognition on the other side
you sort of need representations to get
your theory going
because you need to have predictions you
need to have prediction errors coming
so you need to rely on representational
machinery
so that's what we thought that it's
important for us to
question and think about whether it is

actually the case that
um these two things can be made
compatible
so on the one side the predictive
accounts of cognition which take uh
representations in a very literal
account
um and on the other side the inactivism
uh which uh for the majority of the
people on an
activism side uh representation
cognition doesn't come down or is not
all the way through
representational so this is sort of like
where we are navigating the literature
here
it's sort of like between the very
representational
um predictive accounts of cognition and
um and
the realism that that's a stand that
they kind of need to take and on the
other side the inactivism
which there's sort of like a tendency uh
recently to be made like sort of
compatible under the same roof
so to speak great thank you both for
contextualizing and everybody can raise
their hands so that we can make sure to
hear
from everybody's excitement here um but
yes the bigger
question is does making a model have a
realist
interpretation if you make a model of
the brain as a steam engine is it really
a steam engine
no but then if you make a model of it as

a free energy principle
corollary process theory is it really
that that's kind of the
discussion and there's a lot on the line
which we're gonna tease out so
shannon and then anyone else who raises
their hand
yeah so sort of jumping to the end of
the paper already
you say like the model in scientific
practice is a representation
of the nervous system to the extent that
it holds explanatory capacity
and i think um like a lot of
neuroscientists that i talked to
if you start bringing up you know the
representation or the representation
debate that's like ah that's
something that philosophers worry about
um but we know when we say
representation we just
kind of need a word
stand in for our conversation so we can
talk about
um like patterned activity of neurons
and we don't
care that it could stand separate or not
from
the situation or the environment but
it's useful
it's a useful word to use so that way we
can have language and talk about
the system or the pattern of activity
that we're interested in
that's associated with some behavior um
and i i just think that sounds right
like how you conclude it
and maybe i didn't have a question maybe

this was a moment where i had to comment
more of a question
so apologies but if that sparks some
conversation that would be great
it's something that stayed with you and
will change how you think about
even other kinds of models next time you
see a linear aggression
maybe you're going to wonder if it's
being used in a realist or an
instrumentalist
context so thanks for sharing it shannon
stephen and then anyone else
sort of one thing i think is quite
interesting is the
just like shannon was saying about you
know often a word gets used but then
it's kind of fuzzy and then it kind of
gets like hand waved away around
why it's maybe not that clear i think
this also may be
the same challenge comes up with
pragmatism because pragmatism is really
useful and it gets used in a lot of
context
but now because everything's slightly
pragmatic in the sense that it's all
about action
through the free energy principle it
becomes so cool
instrumentalism sort of helps in a way
get a bit of clarity when it's something
which is about theory so i wonder
i'd be interested to know about how the
pragmatism
um you know wasn't used as the word and
used a more sort of
clear unpacking i suppose

yeah great question so anybody authors
or
otherwise what is the relationship
between pragmatism
and utility and then instrumentalism
so we're using the instruments
ostensibly for pragmatic
ends but the authors chose to frame
the discussion in terms of realism
versus instrumentalism instead of
pragmatism versus say anti-pragmatism or
some other you know could be pragmatism
versus realism perhaps
so any authors or other hands
um i i don't mind saying just a few
words
um and then perhaps i'll end it over to
thomas but i just want to
um the one way that i think that it's
useful for me to think this
is um the use of representation so
in what sense are we using
representations because
um shannon was saying she was raising a
very good point it is very useful to
refer to these things that this activity
that's going on in
in the brain as a representations but i
think that um
what in what sense are using
representations is important because
if it's in a sort of a metaphor then
obviously there is
no problem because it is actually very
useful and we are building all of these
um
we are trying to make sense of all of
these data that we gather from the brain

uh and that's in some in some sense
building a representation because what
is a representation
representation is a is a sentence it's a
picture it's a map
these are kinds of representations why
is this problematic
it's because these are things that we as
human beings do
we do that in our daily lives and we do
that in our labs
so we build these representations to
make sense of uh the observations that
uh that we that we we do in our labs and
in our daily lives so then we can
obviously
use this word this concept but it
doesn't have the same meaning
or it does in the realist sense so i
think it comes down to how serious
are we in the use of these of these
words
are we using this in a literal sense in
the sense of like sentences
maps pictures that we as human beings
have and build and develop and the brain
also has them
or are we using in a metaphorical way
which is if it's in a metaphorical way
because it's useful
to use the concept then it should be
fine it's a metaphor
okay and that's it really interesting a
metaphor
is really an instrument when somebody
says that
this movie was you know was like it's
like an ocean of ideas or it was like an

ocean a simile
they're using that sentence
representation
instrumentally to convey a point to
you know communicate something that is
internal to the thinker and then
that is not meant to be interpreted
realistically
but in science where there's a strong
realist
kick amongst some people they think it
isn't science unless it's not
gonna be the realest most mechanistic
thing now whether it's
mechanism all the way down or what what
the lowest level mechanism is that's
sort of where
it the those realist ideas bottom out
but without going
that way it's just it is across fields
and across different media we ask
whether we're meaning something
really or instrumentally so any other
thoughts what was some like uh we have
oh yeah scott people can raise their
hand on shitsy because i'm not always
looking at the actual video screen so
thanks
scott and then anyone else i was just
wondering it's it's interesting
um in terms of the encoding and the
decoding
notion so if you have different
isms on either side of your encoding and
decoding that seems like where you have
the
problem potentially so someone's
encoding a communication

as a metaphor but it's taken not as a
metaphor but as a
something stronger and is decoded that
way
then then there's a miscommunication
one of the things that i've been trying
to fuss with and it feels like this is
that place
the painter kandinsky said that violent
societies yield
abstract art and i always wondered
whether that's reversible
whether abstraction is a form of
violence
and here that feels like that might be
relevant
because if i abstract something with one
ism
and then it's in my encoding of my
message and then it's decoded by someone
else who's decoding with a different ism
then that abstraction has led to a
violence in fact it might lead to
actual harm to that other person because
they may misinterpret
my message based on that schism of isms
that you have there i had to go there
but um anyways is that something
that that challenge of communication
across
boundaries feels like it's
invoked here and what might be some
strategies
for doing something about that
thanks scott for the great question
thomas and then
we just continue down the stack so
people can just throw up the hand raise

whenever they have a thought and we'll
get to everybody
yeah i think um these these three
comments really
uh together cut at the heart of the
matter
um i think that indeed from what i
have heard myself neuroscientists pretty
much
often know what they mean with
representation
and they don't take it to mean anything
much more robust than a sort of
correlation
between particular patterns of activity
and particular things that we see
happening um and it's more robust than
that but
not anyway what the philosophers take it
to mean
um but the issue is then
exactly in communication
where um it feels like
among neuroscientists it's a bit of an
[Music]
in insiders knowledge
that representation is sort of a
metaphor it's like
the computer metaphor as a whole
um it's a metaphor and you shouldn't
take it
uh in a realist way as we would say
um but when students
read a handbook this is not always that
obvious
and i've also talked to neuroscientists
that
really believed that the rats had a

literal map
in their brain like we have a sort of
jpeg on our computer
um because they were talking about
representational maps in the brain
and if you know a little bit about
philosophy
in this area philosophers take these
new scientists talk about
representations to mean
just about anything that they like um
so i think
that the the issue is indeed one if you
want to talk about that way of encoding
and decoding that
they encode a metaphor and some people
that are
in the know decode a metaphor
but many people don't and
i've noticed that exactly with the free
energy principle
where it feels like it gets very
muddy with what is metaphor and what
isn't and what people take to be
metaphor and what isn't
um and that sense my my main project
here
is is to try and create clarity um
in what it could possibly mean
and even if people completely disagree
with the position
i think at the very least i would like
just people to be more clear about what
they really mean
thanks thomas and really one of the
goals of these
discussions that we're having so we're
going to go to shannon stephen and s

yeah this is actually really great um
thomas and scott
i hadn't thought about it this way
before
so no one in the general public cares
about the representation words
that might not be true but like nobody
cares
but whenever science from a
neuroscientist
gets picked up by a journalist who then
makes the article with the headline that
says
your brain has models inside of it that
it's running and
you should learn music because it will
make you smarter but learning coding
won't make you smarter which
was like a headline recently um
but that translation process that scott
was talking about that encoding and
decoding
really gets stuck
and it doesn't get decoded when it gets
written about
in sort of like more
mainstream news outlets like maybe
someone's medium blog post that they're
writing
and they're a really dedicated grad
student who's grappling with these ideas
all the time and they're very clear
in their writing about what scientists
mean when they're talking about a
representation or a model
what this means for the actual brain
itself and then what this means for you
as a person just reading about it um

but that doesn't always come across in
like the headline on cnn
or or like the the the facebook feed
that's scrolling through
you know my adult parents computer
screen
and so being clear
you have to go through this step of
using it as a useful metaphor in
neuroscience
getting clear what the metaphors could
be if they're real
or just instrumental readings in
the philosophy camp and then like the
one more step is bringing the science
communicators
in and and getting them to to parse this
huge debate down into something that's
able to be distributed to folks who
aren't enmeshed in this every day
thanks shannon so stephen and that's
sasha
and well this clash is definitely um
a useful discussion because i think in
activism radical inactivism and
free energy are both trying to come in
and say it's not all in the head
so just like you're saying this the
challenge you've got is anyone and as
soon as people
talk about neuroscience and that it's
like there's just an assumption in our
culture that's all in the head
it's not even so that metaphor is like a
signpost
and then everything else becomes and is
it is it a film playing in my head and
it's like well wait a second it's the

mind-body environment with the dynamic
system it's about action
and they look at you with this like wait
so
it's like it's like it slips into that
so i think this is a really good
good point because like metaphors when
you embody them
are kind of like you inactively
understand what it's like to be inside
them but then when they just become like
a signpost
for what to do they lose all of that
kind of architecture so i think this is
this is quite valuable
yep the metaphors and the language
stephen and related to the science
communication point
it's uh oh well this is hardwired in or
the dna
codes for this protein it uses the
language
that's implicit in a very specific
understanding
and that metaphor gets over interpreted
and then you start
where's the ram in the brain you know
where's the gpu and then you start
building out this metaphor before you
know it you're way down the rabbit hole
with something that was basically in
terms of this paper
taking a realist reading on an
instrumentalist
project and again we just think about
other kinds of modeling if you did a
steam engine model of the brain it's not
a steam engine if you do a linear

regression it's not a linear regression
so if you do an active inference model
it's not an active inference brain
but that is very nuanced to communicate
because of the way that our language is
so ines and sasha than anyone else
yeah um i think these are very good
remarks i um
just wanted to um i think i have to
disagree just a tiny bit just to make a
point about i i think that there are
real issues and there are real dangers
in conceiving of uh in the ways that we
conceive
things and especially in the ways that
we use our language
and our concepts and particularly
looking at the concept of representation
but there are others that i can think of
i can think of the concept of hierarchy
for example that
could be a metaphor could be used in a
realist um way
uh there there's a there's new papers on
this which are very relevant and very
interesting
but what i want to say is that in the
particular case of the representation i
think there are real dangers
in using it's not just how the science
is communicated or
oh maybe this is not a problem for me to
deal with let's just philosophers handle
that that's not
real for my particular lab it's not
going to be an issue i think it's going
to be an issue
because i think the way that you

conceptualize however the processes are
going on in the brain
is going to dictate the way that you
design and you interpret the results
so i'm thinking for example there's a
bit in the paper where
we do we make a distinction between two
kinds of modeling
techniques that we can use and they have
two
very different goals ambitions worries
and they end up in very different
conclusions
and um so one of them
um takes up as for free that
there is information in the brain so
this kind of modeling
is the kind of modeling that uses
structural causal um
models and uh what they aim for is to
think about uh the the relations between
the
the nodes uh these are nodes that are
highly connected and the way that we
design and understand or interpret the
data coming from
those models is by the presence or
absence of
information now this means that we are
taking information for
free information for existing in the
brain so what kind of information are we
talking about is it the kind of
information that i also have in my model
because if it is then we're talking
about semantic information
and i mean it may be the case that the
brain

actually works with the very same kind
of information that i have in my model
but we need to prove that that's an
assumption
and this is how we are building our
models and i think that's
that's one worry now the other worry is
that in building these models that
that's the inactivist worry
uh which stephen was just mentioning
which is
if we say that well the cognition comes
down to this representation
representations happening in the brain
then so much for
the rest of uh cognition that is uh in
the body in the body being in an
environment so that this old is an
activist
sort of like tradition and i just wanted
to add that
cool thank you um we have sasha shannon
alex
um yeah thank you for explaining that
so well um and bringing this to the
forefront that
this is an issue to be grappled with i
think um from the neuroscience
background
uh yes it is just relegated to uh
philosophers to deal with uh
these bigger questions but um
leading to this uh going back to this
idea of encoding and decoding
it really does matter what is meant by
the phrase
and then how it's um decoded
and i think in the world of like

molecular neuroscience and um just
recording signals from the brain
we really think that um
[Music]
we're measuring something real about the
system whether or not that's what's
actually happening that's just one
measurement
of the overall behavior and
um to just lean into instrumentalism
it's a useful metaphor for what we're
seeing
but um obviously it
people don't have the training or the
decoding of the philosophy perspective
on what that means
and so um
i've never thought about it until this
paper um that this could be a real issue
even in my own work
so i think it's really important to um
get into the nuance of what we mean when
we use these metaphors thanks
sasha shannon alex
well okay yeah go ahead shane um yeah so
that's really great i
i didn't mean to say that like only
science communicators and philosophers
are
should worry about this question um but
i think so like something that that this
reminded me of in
sort of my field so i look at music
perception
um in the brain and i kind of have a
question of does the brain use internal
models
to predict timing mechanisms and

this metaphor that you brought up
that how it affects the assumptions that
you make and how you design or analyze
or interpret your experiment um
i think it also makes a case for
pluralism like if you're realist then
pluralist about realist approaches
if you're making metamorph metaphors
then pluralist about
the type of metaphor that could be
useful so we
thought or a lot of neuroscientists
still think there's a clock mechanism in
the brain
or a pacemaker that's just like tracking
rhythmic activity
but maybe there's this move now that
there's more of a bayesian mechanism
that's building up
a timing model that's now predicting
when
or what the next event will be that
occurs
and one of those might be more
explanatory useful
and the other one will fall away but i
don't know if
we can actually ever prove that there is
a bayesian mechanism in the brain
or that there is a clock model in the
brain i think as far as we can get
is that taking this approach is more
explanatorially useful
than taking the other approach
cool and it made me think from my
background
as an evolutionary biologist about the
question what is a gene

and that's a philosophical question that
they don't really discuss
too much in genetics departments but of
course the way that you consider what a
gene is
it's going to set up what you measure
who's included on the research project
how you write about it how you analyze
it i mean it's the
primary determinant so whether what's
happening with the electrical signals in
the brain
is part of a real model that is being
built out
or whether that is something like we're
just measuring the temperature but then
we're modeling something totally
separate and then also just quickly um
the pluralism it's easier to be
pluralistic
with metaphors because it's like we can
always write new poems we can always
have new metaphors we can always build
and do yes and with the metaphors
but realism it's a little harder to be
pluralist
because ostensibly it's not multiple
things
at once or potentially you know we can't
all be living our own truth for
what the brain is but we of course can
have our own perspectives
so actually it totally opens the door to
pluralism
in a really qualified way where we can
come together to talk about what's
useful
rather than debate about who's right

about how things are
so really nice framing so alex and then
in s
thanks for me as i came from
engineering world to
learning active inference
instrumentalism approach was an only way
starting to understand what what is
happening
uh because uh when i started to for
finding
and look at uh models uh in this
framework it was a way
and to for my mind to get the basics and
trying to
go deeper uh but as for this paper for
me it's great because it gives
great promotion for instrumentalism
approach
and i just think if there is a notion
for
radical instrumentalism
if so can we use it and how it should be
described
uh and again for me
honestly i don't care much about
philosophy debates or even neuroscience
debates
i'm watching on a world around in terms
of systems
and trying to understand these systems
basically as
black boxes understand their behavior
without understanding how this
deep rabbit holes inside working
so that's why
it's interesting for me is for reading
this paper and

with discussions too thanks thanks
alex so ines and then thomas
yeah um i just wanted to add something
to what you were just mentioning
danielle
um about um the the pluralist approach
um i think that yeah you were you said
that something that sounded
very very right to me which is one can
only
take this pluralist or is it at least
easier to take the pluralist
route um if we are on the sort of the
instrumentalist
side um so for example uh with
the example that we have we have in the
paper with these two
kinds of different modeling approaches
that we can take
like dynamical causal modelling or
structural causal modeling
if we take an instrumentalist approach
then we can say that these two
different sorts of modeling techniques
they can work together
where for example what you get from the
structural cause of modeling which
is you get like a sort of a nice
topological description
of the connections arranged in the brain
as we get a nice map of that which can
sort of combined with
dynamical causal modeling where you get
for example the dynamics
in terms of states and how the activity
occurs you can combine this in a
pluralist way as long as you take this
much more instrumentalist approach and

you don't think
you know it to be the case that what
you're getting from
these models is what is actually the
case
so i just wanted to add that yes
like we have one instrument that helps
us find the structure and topology the
connectness we have another instrument
that measures
or helps us do inference on a slightly
different attribute
and it's not a philosophical debate
whether it's actually just dynamics or
actually just topology
it's actually us investigating a system
we care about and then that's what can
bring
people to the table so thomas and then
scott david than anyone else
yeah um i think that's i think that's
really quite
right um i've spoken about this paper
with other people as well and some were
worried
um that instrumentalism was
placing too strong of a limit
on our scientific endeavors
that if we say oh well we could only
ever be instrumentalist
then they would they said
why do science at all if we can't really
get at the heart of the matter
if we can't actually add some sort of
finish line
say okay well now we have just
an accurate description of what
cognition is and

how organisms navigate the environment
um and i think that's a fair worry
um i wonder if perhaps it's just a bit
too ambitious
uh but you know who knows what we can do
in 100 years
and i think i do think in that
perspective it is important to
see that the paper doesn't really get
into
that grander philosophy of science
debate
about model instrumentalism
across the board right so
it could very well be that at some point
we'll get a model that really well
describes everything that
really happens in in an organism
maybe the main point
here as i take it anyway might disagree
of course is that at the very least
in the free energy principle you don't
get that entailment of
because we can model it the organism
models as well um
but it's not it could be that
in the end we can take a realistic
position on
some other model in the future
nice so we're not committed to yet this
instrumentalist across the board
we can be realist elsewhere and we can
be realist in the future but you've laid
out
a research agenda and a logical
structure for asking whether
you know have we met some of these
claims that were

brought up in your paper on another
topic or for free energy
in the future on that note to quickly
get
back to what alex was saying
it doesn't necessarily mean that we have
to commit to radical instrumentalism
we don't have to be that radical
although that might be called
engineering
by some so maybe it already exists so
scott and then in s
yeah and along those lines so i
mentioned before that i was an attorney
so
rhetoric is persuasive speech
and it is what we attorneys produce
and so it's interesting when you talk
about instrumentalism and realism
in terms of rhetoric so
i used to say to my kids all the time
that the only time you see reality is
when you see paradox
and any other time you're living inside
a model
and so that means that when people talk
about resolving paradox what they're
talking about is whose model wins
but i always told my kids you should
manage paradox not try to resolve it
because that is reality so
let me take that what i'm suggesting is
metaphor may be reality in certain
contexts
because the management of the paradox
happens through rhetorical speech the
incommensurables are brought together
with with

with the rhetoric you you create a story
to make it
a thing right and so what i'm wondering
is
i'm i was thinking of the analogy of the
well-tempered clavier when
bach was tempering musical notes right
there's all sorts of notes out there
and then the scale was decided upon and
now it's a scale and it's objective and
it's mathematically demonstrable why
it's
a good scale but it's not the only scale
and so that kind of idea of managing
paradox through i guess it's
instrumentalism because you're allowing
for multiple
approaches what i'm wondering is
is there a preference for one or the
other
um in in trying to
motivate change in in in groups and then
the
second part of that so that's one and
then the other part is
i beginning to understand that the i
believe that the
mind doesn't exist in the brain it
exists in language
and the brain is an antenna that's tuned
into that
so from that perspective can anything be
other than instrumentalist
so you made a claim about what the brain
is so that was a realist claim about it
but i yes great questions scott so we
got an s
blue

um okay i just wanted to um add
something to uh
what thomas was saying actually uh
whether we lose something
when we take the instrumentalist part
and
i think that my concern is that i'm not
sure if we can
if we can um if you can even have the
realist path
and i don't i'm not sure if that's an
option um
so the reason being and i'm going to
come back to the examples that we have
in the paper because i think these are
like
practical you know useful examples are
concrete examples that we can think
about
amongst ourselves even if we come from
very different backgrounds
but uh there are some scientists here
some neuroscientists here and i think
that's useful to you know come back to
uh what we're really talking about uh
with examples
so with the two different kinds of
modeling or
let's say just two major ones that we
refer to in the paper
where uh you have one kind of modeling
that is trying to capture
a topological description of how things
are
and then you have this other kind of
modeling where you're trying to explain
why things behave in the way that they
do right and that's sort of like the

more dynamical systems one
um so in a sense in the in the dynamical
systems one it's
explanatory because you're trying to
explain why things behave in the way
that they do and the crucial point here
that we also point out in the paper
is that when you explain to answer the
question why you need to have a theory
you need to have on a hypothesis
so this means that these models in
dynamical causal modelling they are
hypotheses dependent right so you are
looking for an explanation as opposed to
a description
um in uh for example structural causal
modeling
attracting functional connectivity for
example so what i want to say
is that in these two particular
ways of modeling things you already find
two different uh two different goals and
two different strategies one is
descriptive
the other one is explanatory now my
question is um
whether modeling this uh in this way we
lose any explanatory attraction
i don't think we do i think that this
depends on what we are doing and how we
are doing because if our research is
going to be
driven by a hypothesis then you're going
to get this explanatory traction and let
me just give you a very
quick example so let's say that you are
modeling a pendulum
right you're modeling a pendulum and

you're trying to explain uh so there are
two things you can do with this you can
describe the pendulum and you can also
try to explain come up with an
explanation for why the pendulum behaves
the way that it does
right um and um
once you do the explanations obviously
it's very obvious for us because it's a
very simple
uh system uh the explanation is gonna be
on well that's the laws of physics
obviously right
but you have what you've just done is
you've just
explained what a pendulum is doing why
the pendulum is behaving like that by
using these two different ways of
modeling the pendulum
why it behaves in the way that you do
and how it behaves in the way that you
do you have your why
by going to by using the laws of physics
and
you cannot say you cannot at any point
say
that the pendulum is representing
the model and yet you explain
why the pendulum is behaving the way
they detect that it is
so i just want to say that i'm not so
sure that we should take the realist
or that we need to take the realest part
just on
on the account that perhaps you lose
explanatory attraction well we are
explaining the natural world
by modeling the natural world

so i just wanted to add i'm gonna
retrace
that gem of an argument actually
realism is looking for descriptions
explanations about how things really are
instrumentalism is testing hypotheses
and then the key twist in the discussion
in the argument
is our hypotheses are about
explanations of the system so we
actually
get the benefits really without the
baggage of realism because we're
hypothesis driven investigators as
scientists
searching for explanations through
hypotheses so by centering
the hypothesis quest we can be
instrumentalists
and get the realist fruit but if we act
like we're realists like we're searching
for an explanation or description of a
system
we get lost we're not hypothesis driven
and we're trapped with all this baggage
of realism
so very clear and really thanks for
sharing it we got
blue thomas stephen
so i hate to like jump off of that like
wonderful
um explanation by ines uh like regarding
realism but i have to go back to
uh what scott david just said about the
mind and the brain because it's
something that i think about a lot
and um that i am
pretty deeply involved in but um

language uh cannot be where
the mind exists and i'm not going to say
that i think that the mind exists in the
brain because i don't but
i think that the mind also cannot exist
in language because that just
it totally um you know negates the
experience of someone like helen keller
who didn't acquire language until she
was
you know well into her late childhood so
so before that did she have no mind
like i don't think that that's fair but
also like in any kind of
trauma experience people don't have
language
for the trauma so were you mindless at
that time
or are do you have no mind to
conceptualize the trauma so there are
many experiences
that happen on a very primal and very
emotional level
that cannot be explained or represented
with language
right so so i think that the the mind um
can't exist there and then maybe to tie
this back into
the realist um the realist versus
instrumentalist
like is language real right like so
so i mean i i'm one of these people that
that really questions reality
and what what is real like what can we
take to
to be a realistic representation and so
language to me is an instrument
so i mean it's something that like we

it's it's not even it's it's a human
concept
so it doesn't exist outside of our
perspective reality which i mean to me
like
it totally pushes the realist debate
off the table because like what even is
real
and i have a neuroscience background but
i know like even in neuroscience
there are very wide schisms like people
that think that
reality is a concrete objective thing
but i'm like you know reality exists
here and like that's it
right so so i'm on that other side of
the wall
um anyway that's it so sorry no
questions just some thoughts
thank you thomas stephen and then anyone
else
um i think my initial point was
uh just to react to ines her beautiful
explanation there and i really think
that should have been
in the paper because it's actually a
really really good way to set it up
um yeah that's really good
as for the the where the mind
is um and what's real i think um
i'd love to chat on about that for
probably hours i think it's very
interesting
and i think the inactivist perspective
is a very nice
middle passage between there's an
objective reality uh that we are
approaching

or it's all in the mind um
however i think really getting into that
will probably take us a little bit too
far
away from uh active inference so i'll
i'll leave it at that but uh
yeah the you know the conversations were
on the freeway
all these fun keywords and side
discussions the off-ramp so
thanks for sharing it steven then marco
yeah i mean the culture culture does
come back to
how we frame looking at reality so we
tend to look at reality
um either with a perspective and model
or
to get that perspective we use these
tools because we're a very kind of like
tool based kind of culture now um but i
like in the paper it does say
it says in that first abstract it
remains disputed whether it's
statistical models
are scientific tools but then it says to
describe
non-equilibrium's steady-state systems
so in a way
non-equilibrium steady-state systems is
kind of like a foundational
ground that that that could be to some
extent real
it's the question of the models that are
used to describe
how they work that isn't real
you know so i think that's also like
this doesn't
like there's still room for the idea

that the brain at some level
even though it's quite abstract is a
non-equilibrium steady state system it's
just
that the models the statistical models
shouldn't also be the
i don't know the way that that system's
doing its job so to speak so i think
that's kind of
and i suppose what that also then brings
back is how do we get at that and this
is where
i'm quite interested in indigenous ways
of knowing because well
if you don't come from a perspectival
root and if most of
what if you come from other like
metaphors is one level but other sort of
swarming
kind of um
non-non-literal ways of knowing
you know non-verbal maybe you that
that's another avenue that opens up that
says okay maybe we have to come at
it through other ways of modeling anyway
thanks steven marco than anyone else
hi thanks um so first of all uh just
fair distortion i haven't fully read it
yet
um but um it's an initial discussion and
and uh scott mentioned paradox and i'll
take that up
um i actually suspect that
um the paradoxical interpretation might
actually be the correct interpretation
so we have a nice
um table of possible interpretations for
the fvp

um a two by two uh table but but what if
it's all of them
right so so the fep especially in the
elaborations and active inference
is already kind of paradox because you
already have these trade-offs between
accuracy complexity or
the epistemic and the pragmatic aspects
so in kind of
a weird way the whole discussion can
flip up your other different
interpretations
seems to me to parallel the way that the
let's say first deflationary the
statistical densities in the brain
actually fulfill different roles
depending on the actual relation with
the world
right when i'm actually just imagining
about the world
then the representative relation between
the densities and the targets are not
actually with respect to the world
they're actually respect to each other
in my brain right but nevertheless uh we
still situate in the real world
at the very least if we assume an
objective reality as in a reality that
is not dependent upon our interpretation
until we act upon it then at least we
can say
that by virtue of fep at least what it
suggests
that the way that these models sorry the
way that these statistical structures
and densities are
shaped are shapen by something that is
real

so in that sense i i personally um
sorry i have a lot of allergies for for
some stuff philosophy but
um i i'm very much against this whole
notion that the representational content
or representations
have to be adjudicated by some accuracy
or correctness condition
and i think it's exactly fp that
challenges that
right why the hell would that be
reasonable that a reputation has to be
correct
rather it is that it is shaped by some
sort
of correctness right but if you would be
stuck in that we would never get
somewhere
right as an organism it's not about
being true or wrong to correct all the
time it's about being so ratchet
yourself
in virtue of this adaptive imperative um
i'm not sure where i'm going um i still
have to properly
observe everything but um maybe not
actually type this up is what would you
think
of um instead of choosing one of the
tables
um cells one of these possibilities
more embracing what is in my opinion
applying the fap and saying that it's
actually paradox if it's all of them and
none of them
thanks marco and i also agree the fep
helps us by thinking about how the
observations get to us

whatever side of the blanket or the
interface we're on we can take those
observations with
deathly seriousness instrumentally it's
not that the actual movement of the
water molecules is gonna kill you but
there are temperatures that kill you
and so we can take our measurements
seriously and be
instrumentalist from our side of the
blanket and say yeah you know there
might be a real system let's hypothesize
as instrumentalists about what is out
there
but if we're waiting on our side of the
blanket until we
we have resolved what is quite literally
on the other side of the blanket
then it's going to be arguing about
what's inside of a box that can't be
opened
and then policy failure is going to be
the outcome so it's a really interesting
point about how we're kind of using this
fep scaffold
to actually navigate this paradox so
really cool there so annes
scott thomas
um okay i just wanted to add to uh what
marco was saying i think
it was a yeah it was a wonderful uh
remark
and i just wanted to because that also
allows me to
sort of like draw to highlight a point
that we made in the paper
that was also very fundamental and this
is just that the fep in itself

does not have to entail this table
we can talk about the fep without
talking about this table
and that's a point that we try to make
in the paper so this
table only comes up as something that is
relevant to us
once we start talking about process
theories or the corollary
uh active inference because the fe
p in itself uh comes from uh variational
free energy
which is a measure of the relational
variation of
uh free energy to attain model evidence
in relational base
so we don't have to talk about this
table to talk about
the fep okay and we don't even have to
talk about
active inference or process theories
that's when we start talking about
okay how is it that the system is
minimizing free energy
that's when we come up with these
process theories
to try to explain how is it that the
system minimizes this free energy
right and that's when we start bumping
into these conceptualizations
that um our explanations or our models
that aim to explain the process by which
the systems minimize free energy
also exist in brain
right so we don't have to take any
really standard or any instrumentalist
stand
until we start talking about these

process theories and i think that
sometimes
i think that it happens a lot that in
the in the in the literature these
things come
conflated that the free energy principle
necessarily entails active inference or
necessarily entails or is compatible
with process theories like prediction
error minimization or predictive coding
or protective process
process and these are actually not
there's no entailment
between them as as we saw and as we as
we try to point out in the paper
if i could ask a question quick
follow-up on that actually so
let's say we're in the fep world then
there's all these corollary process
theories that are falsifiable we've
talked about that
um last week and in several other weeks
what is it though that leads someone to
accept or reject or be attracted or
repulsed by the fep
so within the fep we we know how to get
from here to there and you're right
there's no strict entailment
it's actually a totally scientific
empirical question
but how do we get to that fep
understanding that leads us to the
downstream process theories
um okay um i think that are i think that
there are
different goals and then at least for me
it's useful for me to
to to think about it like that so what

we have what we gain
we're talking about fep is that it gives
us
a much more sort of like a grasp or
gets us closer to what things really are
biologically speaking so what we get
is we get these um these bounds between
like
life and death by the form of
entropy so it is this key concept which
is entropy that you do not get
in process theories because precisely
process theories are are
in a different game they are in the game
of uh explaining
uh the processes by which uh cognition
is is is an activity
uh uh is conducted so it's the processes
and the fep
is on a different business which is the
business of looking into stage that's
why it comes much more related to
dynamical systems theory
so that's the point that we tried to
make uh
clear in the paper that it seems to us
that comes
very often conflated in the literature
that there is an entailment i mean
active inference is a corollary of the
free energy principle
so it is it is very important um
precisely to tell us how it is that
systems could be
minimizing free energy
but there is no necessarily entailment
even though if you want to tell a
much more detailed story then active

inference becomes very useful
thank you scott thomas anyone else who
raises their hand
thomas i think you had your hand up
first is this something you want to jump
in on
no you okay you're meter thomas
ah yeah my fault um unless you have
something that's
um not really related to the current
conversation
uh you can go first because um if it's
related if it's related thomas go for it
otherwise
yeah yeah go for it um that i
i noted some comments to a lot of the
things people
said uh um
so regarding the question of what
what we get from accepting the free
energy principle
um i'm honestly not very sure
[Music]
i'm still trying to figure that out and
i'm i'm hoping we can get some
very neat statistical models of
behavioral dynamics
and that we can somehow get a better
grip
on the patterns that we see
um and then
coming back to uh what uh marco said
um if i understood
um the story correctly
by saying that um
you know why not all of the options
simultaneously
right so in some ways the model

is a very neat description in another
way
the brain's model captures something is
based on something real
in the world um i think what happens
there and this is this is something that
i've seen in the literature as well
um i don't remember who wrote about that
but i've seen some people saying oh well
you know it's the best of both worlds
because we get both representationalism
and activism
um and i think the issue with that
is that it assumes
that realism is
a valid position it assumes that
we can talk about such a thing as a
model in the brain in a non-metaphorical
sense
it assumes that there is a model in the
brain
which does or does not latch onto
something real
and i think in that sense um
the way that you often see
people talking in activism and the free
energy principle
together is by assuming realism
and um ignoring the inactive criticisms
of what realism entails um
and it was the second point about
whether we need to
see representations as involving non
evolving correctness conditions
um i think that's a very fair point
but my response to that would be
that sort of comebacks comes back to our
um talk about encoding decoding and

language and being clear in general
um we for
the sake of our paper decided what
we call representations involves
correctness conditions
um so if you have a different notion of
representation
that does not involve correctness
conditions that's a totally different
story
and maybe those types of representations
could be totally fine
if all you mean with representation is a
covariation relation between two
different systems
i'm not going to dispute whether that
exists or not that seems
fine but the problem is that
if you are that liberal with what
representations
mean um everyone's going to use them in
different ways
and you're going to get that
we all use them but nobody really wants
to talk about what they
really mean and that's
pretty pretty famously at least among
philosophers
described by bill ramsey in
a 2007 book representations reconsidered
i think it's called
um so
the reason to to say representations
have correctness conditions is
mostly just to get clear on the
terminology
right so just a signal of this is what
we mean

when we say representation and if your
notion of representation does not have
correctness conditions
um that's something else to talk about
and
that's the another discussion is whether
those are valid or not then
thanks thomas and in an earlier
discussion with um
maxwell ramstad and others we had a
conversation about how part of
representations is that like
they have to be able to be wrong or they
have to be able to be off base otherwise
they can't be on base
we've so yes lots of other interesting
work
um scott if you would like and then
stephen marco
just said something going back to the
earlier conversation
it was when we were talking about the um
the basically the again the ism schism
and the ability to
i'm staying with that and the ability to
see inside another system
and discern what was going on in the
system it reminded me of the holographic
principle
and i started reading a paper another
paper last night on why holography
and it was just kind of an interesting
notion that the markov blankets and the
holographic principle bear
a similarity in there's an opacity
to them and so the internal states
and the perception of those internal
states from outside

may be based on different isms
and that's something that may not ever
be um
bridgeable i guess and it may not matter
we're gonna talk to chris fields about
holographs in march 2021
right so coming up very soon thanks for
this great point
scott steven and marco
yeah just to build on the um
this idea that you go from free energy
principle and then we go to this process
theory so it's almost like at some point
you've got to put dimensions on
what's going on like there's an inside
there's an outside there's
so with active inference there's there's
the idea that the stuff is going to go
via the markov blanket and there's
a way to know and that way of knowing
it's
still cut it's still to some extent
dimensionless because um the
there's there's no direct way to
transmit
the dimensional information it's only
through this entropy
and i know entropy is mentioned in the
paper so i'm kind of interested in this
idea of
when you go from say
the entropy in this kind of um
general sort of non-dimensional sense
it's just
it's just a way of knowing um
and then you've got this kind of high
you talk about this idea of a low
entropy state or high entry

states but you've got this kind of this
is sort of the jump between the two in a
way where there's like how
because with active inference it's the
change in entropy
which gives the way through the back
door
to encode knowledge so
because it's through the change in
entropy
in relation to action in relation to
sensorial data
that you sort of somehow build up a
dimensional
understanding even if that understanding
is only in the mind
maybe not at the organism level so i
suppose
um i i think that may be what i know
there's a new paper you brought out on
skillful performance
has just come out so maybe this is sort
of
opening up these other questions around
entropy and
dimensionality and when you actually go
from modeling changes to
how do these changes
relate to information or knowledge
flowing or being generated
you know from the blanket through to the
generative model and the world yep
stephen
thanks a great point you're getting at
this question which is
how does the dimensionally compressed
information from the world
like a panorama that just presents

itself get enriched through our
experience and internal model
into something that has what you're
talking about is dimensionality
and then it's a good question to ask how
it relates to the
thermostatistical lineage with entropy
and some of these very pervasive
top-level concepts so really well
brought out marco
um uh are you wondering um about about
thomas's
comments on my comments um if you could
elaborate on
first of all the um uh the criticism
from an activism
as he mentioned but but i would love to
hear more about that especially
in relation to um maxwell rancid another
paper on an active tale of two densities
um also about the correctness condition
i understand that you have to choose a
definition of representation
and again i do understand that you have
to respect the lineage in philosophy and
what has
been established as authoritative or
conventional
um but but um i i do
believe that that there is a certain
need i think
to also revamp philosophy because i feel
like there's always this kind of weird
dynamic of like the two clubs in the
school playground
with euphosion science one is constantly
trying to radically revolutionize the
other

but philosophy seems to be so stubborn
in not allowing science to reflect
to revolutionize them right they keep
trying to
the philosophers keep trying to you know
to to to
yank the rigid price out of the shell
but vice versa somehow there seems to be
no path we're not allowed to do that so
so
so for example why not let's this field
of science
philosophy um gank that prior of
representation having to have
correctness conditions out of that shell
because for example you can you can also
just kind of play with it right not to
correct his condition
but uh saying that in the limit it
should
settle into correctness relation with
the target
as if it's kind of fixed point or
attractive point right you can also say
it's not a condition it's not a limit
but it's a factor that contributes to
that thing that we call representation
right so so i think i would love to hear
a discussion about that opening up the
whole role
rather than this essential property the
role of the notion of correctness
in relation to representations as
situated in the world
um and in turn i also want to again
emphasize so
even though that you're right in the
difference between the coroner the many

corollaries
potential corollaries of fep versus fp
itself where fap
constrains a particular way we create
process theories
um and there's another thing that i
think is not often enough
talked about which is the kind of
implied picture of sentient systems for
example
uh from fpp and its corollaries where we
not often enough talk about the
inter-relational
um system the way that the systems
incentive agents or complex systems
relate to each other so it's more
specifically for humans or
us um we also self-model right
and when we're self-modeling we need to
model all the different structures we we
possess
and know what role they play in the big
ecosystem that we are
and so in the same way we can say that
we in our self-modeling we also intuit
or sense
that some of those subsystems have a
very instrumental
right if for example you could say that
the executive or
attentional mechanisms of the prefrontal
cortex
is effectively instrumental and in a way
realistic about itself
but the fact that it has to organize
regulate itself
such that non let's say primary physical
observables

needs to remain a certain bounds that
seems to me instrumental because it
doesn't matter if there really
is some some quantity or some notion of
stability
of feeling good or feeling safe that
doesn't matter
right it's just settled in that way but
at the same time
if it does latch onto something real
cool and maybe you can't elaborate upon
it sorry yeah
yeah it's all it's all good no you're
bringing up okay so point one was
related to
inactive critiques point two was about
correctness conditions moving beyond
them
point three is sentient systems so if we
want a question that's answered let's
just try to trace out the threads so
that we can actually
return to it because they're great
points yeah it's great points marco
is that good or is there more it's just
what's the new ones the third point is
more about
in the process of self modeling we
include the notion or
or implicitly intuitive model of the
notion of
do these things have instrumental role
or more of an epistemic
realism right that that that way more
that these theories are more about
modeling processes of modeling
rather than modeling essences and
properties

yes the relational all the way down type
thinking so thanks marco really
three sides of a very nuanced
discussion and an ongoing questions so
thomas than anyone else who wants to
raise your hand
yeah thanks for that um
so i'm not sure i completely follow
everything
so um if i get it wrong just
interrupt me and um let me know um
so i think you mentioned
um regulation like bodily regulation
as a form of self-modeling right
um so i think that
accepting that bodily regulation
is a form of self-modeling
is assuming the realist position
the there is a clear effect that
we see biological organisms
self-regulate
whether they do that by
exploiting an internal model
is another question entirely
and that's exactly sort of the
distinction between instrumentalism and
realism
that the paper tries to tease apart
so it's almost that when you mentioned
the um modeling
the process of modeling is the
instrumentalist side on marco's
way of phrasing the third part and then
the question about essences is the
realist
question so the movement towards
instrumentalism is related to this
relational

insight so i think
the issue is i didn't exactly get the
relational insight
that's the part that was a little fuzzy
for me and i wasn't sure exactly what it
was getting to yeah marco
where do you see that relational just
because it's something that comes up and
and um it does sort of
cross different idea or debates to
so what does that mean to you or how
does it play out here yeah so
thanks so so often when we when you see
literature on academics or friendship
principle
uh there's always a centerpiece of a
market centerpiece of a certain system
slash agents but the very rare one
figure one
i'm showing figure one just if you want
to refer to this
oh yeah exactly so it's always this but
it's never
this with another one that's like this
it's very rare
let me just see market blankets uh
supplies or
markup lengths of live paper that that
does go into it but
but very little and start the shockingly
like these kind of philosophical papers
um is there actually talk about um how
market blankets interact with each other
and the reflective of the mutual
influence
because in that process of mutual
influence how do you actually
do that right how do you model the other

models that you possess or are
right you're not just it's not just in
terms of properties as such it's also
about the role they have
the bigger inferential ecosystem but i
guess maybe that that requires a leap of
faith in active influence first right
if you already assume that the dynamics
are governed by inferential imperatives
then you also have a small step to
believing that
in modeling each other they also need to
model each other's inferential
contributions
which where we go again back to the
decomposition
of free energy principle uh formulation
in terms of epistemic and pragmatic or
the accuracy of complex
right maybe one mode of engaging with
the world
or incorporating one particular set of
systems
of your total constitution will lead to
more instrumentalist
approach right when you're just um
instrumentally engaging with something
not assuming you have a correct view of
something
when you have a new game or some weird
object you don't know what the hell it
is
you're not gonna engage with it based on
a realist assumption
you're gonna engage with it based on
idea of i'm gonna figure out how to
manipulate it i'm going to find all the
instrumental points of latching onto it

right and so no matter what you you
cannot escape
the the fuzziness or the vagueness or
the fluidity
of these so seeming dichotomies in my
opinion
and that's the relational aspect right
that in terms
of modeling and its many facets you need
to understand the relations of these
different systems as yourself model
especially if you're extremely complex
with humans
playfulness human as a curious
investigator playful insight
relational thinking and a very something
that kind of is personal
and almost distinct from philosophy and
or science
it's the personal which is we're really
studying this and we're people who have
certain
um valences towards different modes so
pretty interesting
um thomas in us
yeah um so i think i think that's a very
interesting
question um and
whether we approach something from an
instrumentalist or realist perspective
um but i think still
that all i mean i might be way off base
here but
from what i hear it sounds like it's all
still
within the realm of realism and only
when you
take us to be

modeling at all
can you say about can you talk start
talking about
whether that modeling is instrumental is
the realist
so the instrumentalist position
that um we defend in the paper is one in
which we
only model when we
literally do so in the world when
you're using your computer ram to start
up a model
or when you're drawing a diagram
of when you're literally making models
in any other sort of
activity that you're doing you're not
modeling
you're doing the activity so
in picking up a cup i am not modeling
the cup
when i'm not in any sense
models are not relevant there i'm
picking up a cup
um and then
i think the other point you made a
little earlier
about the interaction of science and
philosophy and
that it seems that philosophers are
always trying to revolutionize science
and science is not allowed to
revolutionize philosophy
um i think that's a fun point as a
philosopher
um but i think that
i think that with the popularity of
predictive processing
and active inference in philosophy

you see where science really is
revolutionizing philosophy as well
and i think especially in philosophy of
cognitive science
there's always going to be a
bidirectional interplay
um i wouldn't have been sitting here if
it were not for the influence that
science is having on philosophy of
course i'm delving with
into all of this um and then when you're
talking specifically about the notion of
representations
um
if scientific endeavors could make it
extremely plausible to throw away the
correctness conditions for
uh representations then maybe that is
something we need to
accept but the issue is that
i think anyway just at first glance so
representations are
it's a word we use also outside of
science right a picture that i draw as a
representation of whatever it is a fancy
drawing
and there are certain features of the
picture that i drew
that are particularly relevant
in making that picture a representation
of whatever it is i drew
now when you want to use that same word
in science
you might just say well none of that is
important and
we're going to drop some of the very
important features of the picture and
use it very differently in our science

jargon
that's fine but
it's a little difficult to deal with um
when we already have a notion of
representation and what representations
are in um just the regular
dealings in the world but
you could technically say sure you know
we'll call it something else but it's
something that we'll have to
all agree on and then be clear about
what that means or not
thank you thomas so s and then anyone
else so we have like little you know
half an hour or so
left so keep any thoughts prepare them
get ready we can flip around to
different slides in s then markup
um yeah i just wanted to uh i first
wanted to say the
great points uh thomas uh yeah uh that's
precisely what i'm gonna
sort of like uh expand or try to explain
a little bit on
there's there's a a lot that that i'd
like to say but let me just um
go back to the self-modeling point as
the bodily regulation
so i'm really struggling to to
understand
what this means and the reason
for that is because we need to motivate
why
would the body have to model itself
so typically one starts this modeling um
sort of like um endeavor when we do not
have
direct access to things so we wonder

about
um what could be you know causes so i
wonder
why i'm really struggling to understand
what this self
modeling is because uh in terms of body
bodily regulation because what is it
that the
body needs to model really because um
we start modeling we want to model
things when we don't have access to
a true posterior when we do not have the
causes so we need to find the posterior
right but the so the self modeling seems
like a little bit counterintuitive
because
it's it's the self that is right there
so the body is right there so it's
it sounds very counterintuitive to me to
think that
that that should need any kind of like
internal modeling um
in in in the body um
so the other point that i wanted to uh
get back to uh was
marco's point because marco seems to be
like a little bit like worried about
um you know the lack if i understood it
correctly the lack of
um having sort of like a a program or a
framework
that would allow us to get like things
um an explanation there would be like
much more relational
all the way down um and how would we
like link all of this together
and then that's what when we came to to
this slide i think

so i just wanted to say like a few words
on that i want to say that the
okay so that the cool thing about macro
blankets
as we know is that uh well we can apply
them
uh regardless of the scale you can apply
them to any scale
and you can nest them and i think that
this is something that
um marco could be like a little bit
worried about and
perhaps this is not going to be you know
the solution for what it was it was
mentioning
but maybe to get us like stuff in the
direction so
the idea is that um with markov blankets
we um actually understand the relations
between
different aspects and levels of uh
neurophysiology
um as being uh uh together relate
related in a multi-scale kind of system
so basically the idea
is that you can use a markup blanket to
zoom into the level of description that
you want to
investigate but without losing sight
that this particular level
is already embedded in a multi-scale
level uh system right so
then this allows us to uh zoom in
um up and down for example within the
brain and
uh that's a paper that i have um is
going to be out soon
so it allows us to zoom in and to zoom

up
and down in the brain but it also allows
us to zoom uh out to the organism and
then get also the organism
um in the environment so that's the a
nice thing that we can do with the
with the markov blankets where obviously
the internal states and the active
states are going to correspond to the
level of description that we want to
look at
and then obviously we are going to look
at the embeddedness of this system
within its environment because its
environment is going to be the external
states
and the sensory states so that's the
cool thing that we can do with markov
blankets is that
it allows us to zoom in and out um um
and up and down in a multi-scale uh
system
thanks yes great points and i think for
this zooming in and out
seems very relevant to a lot of our work
and a lot of what's in the cutting edge
so maybe we can have like a workshop or
a
panel or something like that or some
we'll see it later but this is really
interesting stuff
so again maybe you know 20 30 minutes so
if anyone has
live chat comments marco then anyone
else who uh
then stephen and then anyone else who
raises their hand and then scott
thanks um so she want to go back to uh

models and multiple levels of skills and
stuff so
so um uh thomas says something about
you're not modeling
uh something you're just doing right um
and i agree
again like i said uh my my claim is that
that that it
varies over the situation context so
again that's why i argue that you need
to zoom out to taking the relation
so if you take the paper um on active
infinite curiosity with the
with the games right so first you're
exploring you're just doing the game
you're kind of allowing the statistical
patterns as are impinged upon you
to kind of accumulate shape and adjust
each other etc
and indeed at that point you're just
doing and you're just at the emergency
co-variation of relations etc but at a
certain point
there's some kind of transition you have
a moment of insight
the model sorry the let's say
the picture more naive because in sense
of the game kind of arises into your
awareness as
a complete picture and at that point
you're able
to at least hypothetically write down
the model that you think that the game
is governed by
right and so you said something like
we're only modeling when we're doing the
computer thing
and claiming that this is going to be a

rotational model of the environment
but the thing is what what's the chicken
in the egg
right if you're doing that game you're
at the mercy of the patterns
and then you obtain this internal
understanding of how the model looks
that allows you
to create a computational model that
means that the point where the model is
substantiated is actually
in the agent itself right and so that's
why i talk about the relational thing
and the and
the sound regulation when you add a
system are able to model yourself
then you have different orders of that
model the moment that it's just
intuition just you're guided by the
variational patterns
just letting that be latched onto your
actions and policy selection
yeah it's not really maybe just maybe
it's not the model then but the moment
that
at this level above you're able to say
that this
is a model of that and this is the the
systemic relation pragmatic relation
then i think we should be able to say
that's proper
and model proper right and another thing
so something
thomas said about uh when science allows
us to throw away the correctness
condition
i think we have that in psychology and
in social as

social psychology a lot of people can
develop an understanding or a model or a
stage of the world
that is actually not governed by correct
conditions if you i don't get political
but you know
you see in certain political areas where
people seem to be governed not by
truth conditions but more about
pragmatic conditions right
but which beliefs if adopted or which
ideologies if adopted
will i be better off in the world right
that is not a correctness condition
that's
that is taking instrumentalist or
pragmatic primacy for the sake of
your fundamental imperative of autocracy
sort
self-preservation etc um
and so and and and and uh in this
uh point about non-direct access so yeah
we're oneself but it's most level motor
skill
so there remains inferential distance
between the locus of certain areas of
yourself and other loca
of yourself so the hypothalamus is
crucial in in self-regulation in terms
of guiding the
endocrine system but obviously it's
distant from the actual systems
that is supposed in relation to right so
there it needs to be an acknowledgement
of the inferential relations and
infrastructure distances
by which the self as a whole learns to
regulate itself

um uh yeah i just
leave it at that great thank you so in
the last little bit let's
think about tying threads up and
phrasing things as
what we wanted no it's great marco you
did good
and exploring what we want to do next
week so we're raising questions to think
about
and for next week as we tie up some of
these really awesome
threads so stephen scott
in s thomas
nice um so
well i just wanted to sort of add on to
this um
or sort of maybe this is something that
could come back to um
this idea of representation i suppose it
is ultimately
seen as something that's in the brain
and a model is also sometimes
this is maybe where we get a problem is
we kind of have this
folk psychology that models are in the
brain as well
so if i think if we were to say models
are in our niche so if we say we take
this figure one
the model is something we create in our
external states in our niche that we
work with and i have tools and we
manipulate out there
and we engage with our active state
sensory states internal states to work
with
um it can get rid of uh

it helps to sort of um get around some
of these
challenges um and i think because i
think that that is
um the nature i think this is really
useful thinking about correctness and
accuracy of a model
is when can that
need for directness and accuracy be
um loosen someone um
because it might be that you don't have
to model the body doesn't have to model
its active states or model its sensory
states or
model those exactly but it's somehow
using something like them
and there's some sort of bigger model
which we can try to use as a tool to
understand what that aggregates to so i
think
this is actually maybe opening up quite
a
rich thread thanks
steven scott and s thomas shannon i
unraised skip excuse me thank you scott
and
s thomas shannon
okay so um i just um i think that uh
it's important the question that um
marco was rating so i just wanted to
sort of highlight that and just like
sort of gesture to it for the next
session
um so i think that uh marco was
mentioning um
this important self-regulation aspect um
and i wonder and this is like a very
embodied kind of thing it's not you know

available to our conscious level so we
don't know what's happening so it's not
available to conscious
consciousness and i was trying to
dispute that idea that it needs to be
to entail some kind of like modeling
aspects because for us to you know
for you to tell modeling aspects it's
because there is not a direct access to
the body
and i think that one follow-up question
to that would be to think about
what self-modeling would mean on a
conscious level
do we ever do that do we need to
self-model on a conscious level
right because if we do then perhaps
there is like a sort of like a barrier
between
the self and what i think so it's kind
of weird so i would
really like to discuss that um with
everyone and
um yeah and the other question that i
think that would be nice to discuss next
week is to actually get down to the
nitty-gritty
to the detailed technical details um
that would allow us
potentially to think these things are
what thomas and i had in mind in the
paper
which is whether from the fact that
we can be trained uh to use for example
variational uh
base to uh attain likelihood
of uh model evidence which is like what
we do vision inference and modeling

that kind of thing and also from the
fact that
we as human beings in our daily life we
can think and engage in sort of like
abductive reasoning to
wonder and make inferences about what
what we think is the most likely case of
things around us
from the fact that we as scientists can
do that and we as you know
regular people in daily life can do that
does it follow that the nervous system
itself
or the body itself does that and engage
in those kind of like tools
so those would be mine yeah nice i'm
already capturing them on a slide so
we're definitely going to get to them
ines
thomas shannon marco oh sorry yeah
thomas shannon market yeah thanks
um
[Music]
i have a lot in answer uh to what marco
was saying so
i'll try to slim that down
um as a very
quick probably not to um
convince you but just as a description
of how i would see things
um if you're playing a game and at some
point you figured out
the patterns of the game and you're able
to make a model of it
i would say where the model is
instantiated
is when you draw the model out
that drawing you made the description of

the patterns
is the model and when you're producing
that drawing that you can share with
other people that other people have
access to
you are modeling um
any other modeling models are never in
the brain
models are never in the head models
are in the world
in our interaction with the world and
only exist because
of our social cultural heritage in which
we have learned to use models
sort of what ines was getting to as well
um and then when you say when you're
talking about
some people being led not by correctness
conditions
um i think that's that's very fair and i
think
none of us are generally led by
correctness conditions in most of what
we do except for science when we
give it our best shot um
we're just doing stuff but the most
important thing there
is not whether that means then that we
need to
change representation for it to mean
something not involving correctness
conditions
but maybe that means that the notion of
representation
isn't really applicable here and
instead of changing the words we use we
just use a word
that is more appropriate to what people

are really latching on to
and i think that'll be just
communicatively maybe an easier strategy
instead of continuously changing the
word to fit
our current understanding and just sort
of come up with a different word that
gets better
some deo logisms and also our growing
terms list
shannon then
um thanks for all of that discussion
i'm wondering if in search of terms that
would better fit our description is the
term
expectation more
or less vague than saying representation
or a model and what i'm thinking of
is when you're standing your body has a
certain expectation that you'll remain
standing
if you're standing in a room you expect
like you at a conscious level maybe
expect to stay standing but also
um you know your cerebellum and muscles
and your trunk are expecting to
be in a certain relation to each other
because the ground's not moving
so they're just maintaining you know
some standard standing posture maybe a
little bit of
balance side to side if you
feel like you're standing still on a
train and then the train starts moving
when your expectation was to stand up
straight now you've
have a deviation from that expectation
your cerebellum your motor cortex all of

the muscles in your trunk and your legs
react or
predict that they should have been
standing upright
predict that they should have been in a
certain relation to each other stand up
right
and they either update that prediction
or they act to change the world so in
if it acting to change the world then
all of your muscles will
tense up in a different way to keep
yourself standing upright
if they're just accepting the prediction
that you shouldn't be standing already
you'll fall over and i just wonder
is it helpful to say that your body has
a model of standing
or like an s says your body just is in
direct access to itself
so even if there's an expectation of
standing upright
there's still
like maybe it doesn't help to call it a
model but there could be correctness
conditions in that case if you called it
a model and
like if you're so far out if you're so
far out of allostasis
right then you're in an incorrect
correct correctness condition for
standing
and so to respond to that perturbation
is to bring it back to that
that correctness condition and i don't
know i don't know
i guess does expectation help or is that
just even more vague

than saying that you have your body has
a model of itself
fun thanks shannon uh marco and s
yeah actually um what channel says very
nice because
i think again that re-emphasizes the
importance of
of distinguishing the kind of
operational constituents
of how models play a role in agents so
what she was describing seems to me like
the interaction
of of um how the uh
afferent and efferent expectations and
data have to interact and that indeed i
agree that that doesn't have to be
called a model
i would just see that as um the element
or the operation
of inferential interactions right and
and kind of a zeroth order or a proto
model maybe
but again um it's on top of that when
you have the
kind of order above that where i think
you would have to say that
there is a model because then there's
also the modeling of how these infrared
constituents
relate to the world that has the model
be modeled too
uh for example sometimes we interact
with the world
in weird ways and we notice it and we
have to actively reflect about how
we were guided by our actions i mean
i don't know which philosopher was but
you know philosophy about self

examination
examining your beliefs assumption if
that's not modeling i don't know what it
is
um and and um he has raised some very
nice points
um so the rule of consciousness uh
too about conscious of regulation and i
i personally believe
i am one of those people does attribute
a function to consciousness or at least
attribute a functional drive that led to
the emergence of consciousness
where by the sheer fact that it is
uh the kind of the apex of higher and
higher levels of modeling
similarly how multi-sensory integration
is a consequence of this
in the very same way um it seems
it seems stands to reason especially if
you look at works like graziano's
attention schema modeling that's the
very
drive that led to motorcycle integration
and consciousness
is the need to model that which is below
that um i'm sure there's some there's
some activation papers for that too
um i forgot in response to what it was
but also wrote down somewhere
that um it's relevant to take also mel's
paper in this discussion about models
because i think what's very beautiful
there in contrast to this discussion is
the
diversity of nuances regarding what a
model can be and what they are useful
right although that would probably stand

at us with a lot of inactivists but you
can draw a lot of parallels between the
different roles and values that model
could play and have
for a scientist and the various ways
that models as
embodied could be useful right for
example some are
good for for conceptual exploration some
are
more truth-oriented some are more to
kind of integrate different things and i
think these are all valid ways to look
at models
um and they survive simply by virtue
again by virtue of what free energy
principle entails is that
the the success of these models is what
leads them to persist right and the
success
in terms of their pragmatic and
epistemic contributions
to minimizing or reducing variation free
energy
right and so it's like oh wait yeah alex
uh john says
a recent paper right or well one of them
had the recent paper on it's not about
it it's about
uh what must be true if you've gotten
there actually right so what must be
true
yes yeah yeah x constant so so given
that we have
bernoulli assume models what must be
true uh given that we have them
right so the answer would be the uh they
got there because apparently they

contributed
to a reduction of variation free energy
right
um also i thomas i gotta say i think
it's it's kind of one of those uh
disagree
uh agree to disagree things because i i
i i feel a sense a bit of an axiomatic
commitment to models simply cannot be
embodied
um although i would love to continue
that discussion and i also
think that adapting a language is key
and again taking the the norm of
principle free energy
it's the philosophers who disagree most
people use it like the way that
i've been describing right folks
psychology and scientists use
representation
in less of a seemingly anal status
rigorous notion of correctness condition
it is simply a part of a bigger picture
of
how their value should be assessed so i
i and i think in general it's a it's
example of niche construction we are
embedded in a world of language we're
our linguistic bodies
so adapting a language instead of
throwing it away seems to me a far more
reasonable approach
um especially if you want to confu if
you want to avoid confusion so yeah i'll
leave it
well it's always interesting to
re-listen to one's live stream
because while we're speaking we think a

certain way
it's a different mode and um and s you
really
have a finesse with constructing
philosophical arguments on the fly
as many other people embody and the way
that that is done
performatively stephen i'm sure also you
know resonates but like
the way it's done for the first time
performatively is very different than
hearing it
in a different state of mind it's like
it lines up
in the moment but then a different
moment and then
it's like because that's so it's really
interesting points like are we self
modeling
when you're improv when you're
improvising are you self-modeling or are
you doing are you just controlling your
larynx
where is consciousness fitting into all
of that those are really great questions
in s and then any like closing thoughts
we could go
anyone else who wants to speak in us and
then feel free to drop off whenever you
want
um okay i just wanted to sort of like
address
uh marcos and then address uh shannon um
point very good points um so there was
one thing that marcus said that uh
strike me very very interesting because
you were saying um about that we need to
so you need to choose the models the

models that are more um you know
suitable
um and i think that um by saying that
obviously we're taking a realist
a a realist perspective as in thinking
that there are actually those models in
the body or in the brain
so that's sort of like what thomas and i
challenge so i just want to
sort of like give my two cents on that
particular thought and say that
i don't think the success in is choosing
um in choosing models
but it's more like the success is in
choosing
the most desired states that are going
to get you what you want which is like
you know
avoidance of dissipation entropy
uncertainty and all of that
um so states are preferable models don't
exist
so that's kind of like my take on that
and then i just wanted to also address
what shannon was saying which was really
interesting because
i thought that it seemed to me that
channel was trying to you know
build up like in okay let's just really
get down to hands-on and see like
how do we really apply these things that
we are discussing here
into like for example my lab um and i
really like that
um one thing that i think that is
interesting that happens um uh
that i see happening a lot with my
conversations is that

we don't even see how we are already
sort of like in completely and fully
enculturated
in a way that we speak about the body
and the brain and cognition
in general with so many assumptions that
these things exist because it's just the
way that we've been enculturated
um so as uh shannon was describing oh i
think that wonder if maybe its
expectation is a better word right
and then like it strike me because i was
thinking about it
okay um so the body has got this
expectation but
from what from what
so is that is there like because for the
body to have a certain sort of
expectation i would probably think that
maybe the body would need to be like
sort of like separated from the brain so
it had what would have like some kind of
like
expectation um or
um uh some kind of like inbuilt system
where the
the body becomes like sort of a
humunculus and also has its own
expectations so
detached i i i was like a little bit
like i'm not sure uh
what to do um with that um
so um yeah so that that would be like a
point on
on whether the body's got this
expectation and um
and um and and and how i think that it
another way to think that would be to

think that
the whole organism is evolving towards
something
right there's a sort of like at least
like one goal that is
shared by all living systems right of
course
interactions are very idiosyncratic
we're really context dependent but
there's like one goal there's like sort
of like
in general times shared by all living
systems is that they don't want to die
so the the
so it is evolving towards uh sort of
like avoiding this dissipation
so whatever whatever the state the body
is at
it's it's got to it's got to be dealing
with uncertainty right
either so then shannon was giving this
the example of like oh
it's standing so so is it there's an
expectation of like a certain
expectation regardless to
in regard to being um standing i think
it's more of like the state
that the the organism itself is
which is either certain or uncertain
it's stable or is not stable
it's it's it's in in in it's it's it's
fulfilling sort of like
a a state that is going to you know lead
me to to the next
stage so i think that would be like a a
a sort of
better way to describe the behavior
um that is entailed in in yeah in this

cognitive activities that we do
because i'm not sure how to think like
the body itself expecting something but
maybe i'm just i i missed
um i missed something here
bill thanks so much for that shannon
marco and then
anyone else who wants to give the last
thought
great thanks for that ness i think um
where you sort of ended where you have
like most organisms have an expectation
of being in a more stable state
than a more unstable state depending on
the context and depending on
the environment and everything um but
the point you made about enculturation
like how does the body have an
expectations
um and i guess when i said the body i
kind of
was implicitly including the the brain
so if when your body's actually upright
your brain and
your muscular system have an expectation
of being stable in that upright
position and i think enculturation um
we could also just talk about
um like the development of
certain structures or expectations or
situatedness or something
so the same way like enculturation gives
us an expectation for a certain
form of language our body just by
existing in the world and being in
standing states in lots of different
contexts
develops an expectation of what a stable

standing state would be like
um but i think where you ended sort of
on
this like constant evolution sort of
away from death or away from
perturbation or towards stability i
think that
that covers and that was helpful
thank you shane marco and then any other
comments
thanks um regarding anticipation in the
body um
some examples i'm not sure if this
translate is normally dutch but
you know if you've heard of it or or
pork
soup is eaten with a
spoon but most people will say fork
especially when you're kids in
kindergarten
so um that is more of a linguistic
example because it relies on
a notion of rhyme which you feel to it
and the prediction error is again
contingent upon the model or the
linguistic policy you have
um but also just in in general um
for example with brain relations or
i don't know imagine you're at a bar
someone really drunk comes to you
and there's a really big guy with scary
tattoos and there's no words need to be
exchanged but if he comes really close
to you
grabs you tightly in the shoulder
there's gonna be expectations and it's
felt in the body
and a lot of these expectations

especially imagine you're a racist you
know and and
and someone colored comes to you so then
you will feel a
physical response of anticipating a
certain action or something
but because it's a contingent of
uncertain racist beliefs which in
culture
there clearly needs to be a certain
interrelation between the models with
the
abstractions that you've gathered
through culture and
the physically or viscerally felt
anticipation of an action
and so i think in a lot of ways what i'm
trying to say is that a lot of the
physically embodied anticipations
clearly are contingent upon abstract or
model or representations of something
because if not we wouldn't have these
visceral bodies so i think a lot of
my personal qualms and activism is often
they look at the manifestations the tip
of the iceberg the marks
the labels but but not the necessary
path that it has to have taken to get to
that manifestation um
and you said also about you know we
don't need models we just get to our
preferred states
but then you get the problem of how do
you get to your preferred states it's
not that easy i want a million dollars
and a happy family but
you know how do i get there and so it's
gonna be contingent on a lot of

complicated detours and i simply don't
believe that you can escape the notion
of a model
if you let go of the whole um correct
this condition but i i don't think you
can escape
at least a model in the broadest sense
as nell's paper i think nicely alludes
to
especially with these examples of
getting to preferred states given the
complexity of our worlds and given the
cultured reality of that experience
thanks mark so i would love to talk
about that next time yeah awesome well
on discord or however we can continue
some text brewing and thinking and it's
awesome because it's just like a
snapshot
into our learning and thought on all
these very important questions
so stephen scott in us
thanks well i suppose one thing in this
mention there and this is
this is quite rich is maybe how does
expectations can expectations always be
and to some extent they can't by
minimizing uncertainty
it's like a backdoor way to get is it
you know so it's kind of
um but i do think there's something
quite interesting about
um another way of thinking about
representations
beyond the model and beyond
some internal brain state and that's
where not but there's something called
mental space psychology which is quite

interesting about
how we might be constructing and
designing
um understandings of a space now i'm not
going to mention that now but that that
just
something that comes up is is there's
maybe there's
other mechanisms for modeling that we
work
with which is in between
just the body moving and it being in the
brain but
is is everything tied to this active
state sensory states
relationship so it has to be capitulated
at some level
using our sensor and using our actions
so there has so anything that's
represented in a way we think it
is somehow being capitulated in this
active inference process it's not
being just happened offline
so to speak without any kind of
embodiment yeah thanks
thanks stephen and the good news is you
only have to mention the keyword wants
for the ai
so scott and then in s
one thing i was thinking about is you
know we talked about this as a
way uh method of analysis of existing
systems biological systems etc
and it's also uh the thing that's
what i've been doing historically is
creating synthetic markov blankets
and i think i alluded that that before a
contract is essentially a synthetic

markov blanket
that creates a being pursuant to the
attributes in the contract and so you're
taking two different systems
and linking them harmonically coupling
them through a set of expectations
that are documented in the contract and
so one of the things that's kind of
interesting to me is that idea of
realism versus instrumentalism
i think it goes away in the context of
synthetic
systems because you're essentially
instrumenting
a system from whole cloth saying
hey we're going to correct create this
entity
and it's going to have these attributes
like a corporation doesn't exist except
through the corporate law and the
attributes that are assigned to it by
shared meaning but it's all fiction but
it's a very broadly shared fiction
and so it's interesting that notion of
what we use as descriptors
for existing living forms
and those things that we set up to
create
niches in a sense for the new synthetic
forms and that's something that
i find intriguing with this model from
the paper
is that it's when it's taken to the
synthesis
of new forms it changes i think it
changes that analysis because it's not a
question of discovery but of creativity
in terms of the setting up

of the attributes in the markov blanket
thanks
scott if i could rephrase that it'd be
like the realist is focused on those
essences of the world they're going to
try and describe what's already out
there that's like big databases of our
current
legal systems but once we're in
instrument landia
we're talking about designing apparently
arbitrary
fictions and then manifesting them and
enacting them
and so there the rate limiting step once
you're taking the instrumentalist turn
is literally the innovation and the
narrative build up and how much
collaboration you can achieve
pragmatically without getting into the
realist allowing people to have a
different take on what is quote really
happening
so really interesting and it may allow
for the design
of scale-free interventions in that case
right because you're not just relying on
what happened
naturally but rather you can create some
predictive possibilities
by the way you design one level to
allow for the engagement on a scale
freeway thanks
all right ns and then all the last
comment
okay so i know that we are already a
little bit over time and i do have
another meeting so i'm just going to try

to be like super concise
and i just want to say like one final
remark regarding
regarding the realism instrumentalism
so uh i just want to say that um imagine
that i have a model
i just have built the model to
understand or make sense of brain data
for example that i collected from the
scan
um i can have two this is a
philosophical attitude
i can have either one or the other i can
go and say no
my model uh but that uses all these
tools
all these mathematical and conceptual
tools um is actually
the how the brain is working and
generating that data it's not only on my
model it's actually how it is
or i can say that no this is actually my
tools that we as a scientific community
have built
to make sense of the data that we
actually have access to
and try to generate the the the activity
that we don't have access to so we need
to infer it
right so just one comment on that
particular distinction
and then another thing that i wanted to
um get to is that uh marco was saying
that
oh i can't i can't see how we can how we
don't have models
how to get from one state to the other
and that kind of thing i just want to

say that in that regard
that we don't have models we do have
representations on the agent level
i mean we think we use abductive reason
we have we we use logics we have
concepts
we do make representations we represent
things we can we can do that on the
agent level
so scientists do it we in our daily
lives do that
um and that is totally fine no one is
disputing that
the dispute is precisely on whether the
underlying mechanisms of
what allows for us to do that such as
the body in the environment with the
nervous system
actually themselves also have that
capacity
so that's the question about models
models on the agent level
yes on the underlying mechanisms of what
allows us to model
that's a different story that i'm not
convinced about
um and then about how do we get like
from one state to the other
i think that we choose
we think we use again uh reason
abductive reason
uh we have representation we also use
all of these intellectual machinery that
we can
and have because we are enculturated
systems well
and um and living beings read language
and we choose how we get from one state

to the next
and then that can be all mapped out with
you know dynamical causal models
if we want to so i just wanted to finish
with that thanks so much
thanks everybody for this really fun
discussion i'm glad that we
opened so many great threads in point
one i'm gonna provide a technical wish
anyone can drop off they want to provide
a technical wish and then a
concluding thought so the one technical
one is it'd be fun to have an automated
system
for philosophical classification of
statements even if only within the
domain specific fep
so a statement like the brain has to
maximize the a matrix
is a really if the brain is it
is a policy selector or something like
that is a realist
take even if the person has some other
perspective but in that sentence
grammatically
versus we use this version of spm to
model this it's like it's an
instrumentalist
sentence so classifying sentences and
then even just like kind of like a time
series like shading our
live stream by the way that people are
speaking with an underlying generative
model and then are they speaking
from a position of implicitly like are
people speaking from a position of um
you know pronouns in one way or in some
other way

these are things that are like emitted
states that could be inferred
from linguistic patterns and then the
concluding um thought was
scott david talk about paradox he's
talked about it before steven brought up
that humans are like a tool culture
everybody not just
digital culture humans are like a
handiwork tool
people and so thomas said what do we
lose with instrumentalism and so that's
really just to conclude by saying
that's what's on the table is do we just
cut our ties with reality
do we fight for realism and whatever
that is lost along the way there
or do we cut our ties with reality and
go fully instrumentalist where does that
leave us adrift
if we have a really good ship where do
we steer it because now we're going to
be in more direct contact
with our preferences with our governance
structures because we're going to be
really clear
that science isn't like advancing some
noble truth agenda
per se it's something a lot more cns
thank you
it's something that's more on the um
utilitarians the technological it's an
engineering
development if the fep is within the
realm of philosophy and engineering but
not
academia for example it's not too
foreign of a world that could exist

which would be different for an idea
than other kinds of ideas
and then it's the heart of the matter
and is the heart of the matter how
things are or is it how things are
related to each other
so that's which story we're in at that
top level is the heart of the matter
that
instrumentalism sidestepping the
question you're losing everything
because the heart of the matter is how
things essentially are well
instrumentalism is only sidestepping
that target of what the heart of things
are if you think that's
essence but if you think that the heart
of how things are is relational
then maybe you can be a realist about
instrumentalism you say it's really
i'm really making this scientific model
it's really the code that i put on the
paper
this is really the conversation we're
happy happening and all these sort of
things like that so just it was really
an
awesome discussion much appreciated all
of you active infra
lobbers and uh we hope to see other
people participate in the future if
you're listening this far
you probably could come on one of these
conversations
hopefully so thanks for participating
everyone
and the forum is in your calendar event
for those who participated live everyone

else just stay in touch but
see you next time for 15.2